RecruitO · Learning Series (v2)

# The big picture, and every tool explained

Start here. This updated edition reflects every change made on 27 October 2026: real SMTP email, manager-driven scheduling, a public Cloudflare tunnel, LiveKit video, AssemblyAI transcription, and the new password-reset and split interview-room flows.

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Updated study guide · reflects the project as of 27 October 2026

# What changed since v1

> **Read this first**
> This guide was rebuilt after a round of upgrades. Boxes like this one flag the new or changed parts.

- Email now goes through a real **SMTP** server (Mailpit removed).
- Google Calendar removed — interviews are **scheduled by the hiring manager** by hand.
- The Gemini **model names were corrected** to ones the API key actually serves.
- The app is reachable publicly through a **Cloudflare Tunnel**, behind a single origin (nginx serves the site and proxies the API under `/api`).
- Interview **video now uses LiveKit** (hosted TURN/SFU) so calls connect across different networks.
- Interview **transcription now uses AssemblyAI** (with ffmpeg transcoding), not Gemini.
- Sessions last **8 hours** and expired ones auto-redirect to sign in.
- Added a **forgot/reset password** flow (email link).
- The interview room is split into a **consent lobby** and a separate **meeting-room** page with icon controls.

# 1. What RecruitO is

RecruitO is an **AI-assisted recruitment platform**. It helps a hiring team screen CVs, rank candidates against a job's requirements, schedule interviews, run a browser video interview with optional live transcription, and reach an evidence-backed decision. Every AI output is explainable, and a **human always makes the final call**.

# 2. The running pieces (containers)

One command (`docker compose up --build`) starts these containers:

| Container | What it is | Port |
|---|---|---|
| **frontend** | nginx. Serves the static site AND reverse-proxies the API under /api (one origin). | 5500 |
| **api** | The Python FastAPI backend + background worker. Runs DB migrations on start. | 8000 |
| **db** | PostgreSQL (with pgvector). | 5432 |
| **blogger-mcp** | Publishes approved jobs to Blogger by email (F11). | — |
| **cloudflared** | Opens an outbound Cloudflare tunnel → a public https://...trycloudflare.com link. | — |

> **Single origin + public access**
> Because a quick tunnel exposes only one address, nginx now serves the site and the API on the **same origin**: the browser calls `/api/...` and nginx forwards it to the backend. This removes cross-origin cookie/CORS headaches and means one Cloudflare link reaches the whole app. cloudflared only makes an outbound connection, so your home IP and router are never exposed.

# 3. The request lifecycle

- **Browser** — JavaScript (api.js) sends e.g. `PUT /api/applications/42/interview` with the session cookie (same origin).

- **nginx** — serves the page for `/` paths; for `/api/` it strips the prefix and proxies to the backend (also upgrading WebSockets).
- **FastAPI** — matches the route, runs the auth guard (cookie → role), validates the JSON (Pydantic).
- **The function** — reads/writes PostgreSQL via SQLAlchemy and queues any emails.
- **Background worker** — later drains queues (emails via SMTP, CV scoring, analysis, manager notifications).
- **Response** — JSON travels back; api.js updates the page.

# 4. Tool glossary

## Backend

| Tool | What it is / why RecruitO uses it |
|---|---|
| **Python / FastAPI** | The backend language and web framework (functions become HTTP endpoints, with auto docs at /api/docs). |
| **Uvicorn** | The server that runs FastAPI on port 8000. |
| **Pydantic** | Validates request/response shapes ('schemas'). |
| **SQLAlchemy** | The ORM: Python classes ('models') map to database tables. |
| **Alembic** | Database migrations; 'alembic upgrade head' runs automatically on startup. |
| **PostgreSQL / pgvector** | The database; pgvector stores embeddings for policy retrieval. |
| **JWT (PyJWT) / pwdlib(argon2)** | Signed session tokens in an HttpOnly cookie; argon2 password hashing. |
| **ffmpeg** | NEW: a command-line audio/video tool; transcodes the browser's webm audio to WAV before transcription. |

## AI & media services (external)

| Tool | What it is / why RecruitO uses it |
|---|---|
| **Google Gemini** | LLM for job drafting, CV scoring, the chatbots, and interview assessment (via the google-genai SDK). NO LONGER used for transcription. |
| **LangGraph** | Builds multi-step AI workflows as a graph of nodes with retry edges. |
| **AssemblyAI** | NEW: dedicated speech-to-text service that transcribes the interview audio (replaced Gemini for this). |
| **LiveKit Cloud** | NEW: hosted WebRTC (SFU + TURN). The interview video/audio relays through LiveKit so it connects across any network without exposing your IP. |

## Delivery & web

| Tool | What it is / why RecruitO uses it |
|---|---|
| **Cloudflare Tunnel (cloudflared)** | NEW: exposes the app on a public HTTPS URL via an outbound tunnel — no port-forwarding, no IP exposure. |
| **nginx** | Serves static files, proxies /api, and sets security headers (CSP, Cache-Control: no-cache). |
| **WebRTC / WebSocket** | WebRTC = peer media (now via LiveKit); WebSocket = presence/signalling + streaming audio to the transcript endpoint. |
| **SMTP** | Sends application, status, interview, and password-reset emails (real Gmail in the current setup). |
| **Docker / Compose** | Containerises and starts everything together. |

# 5. Repository map (updated)

```
RecruitO/
  docker-compose.yml        # db, api, frontend, blogger-mcp, cloudflared
  .env                      # DB, SMTP, Gemini, ASSEMBLYAI_API_KEY, LIVEKIT_*, JWT
  backend/
    Dockerfile              # installs ffmpeg + Python deps
    migrations/versions/    # incl. f1a2b3c4d5e6 create_password_reset_tokens
    app/
      main.py               # app, routers, worker, /health
      dependencies.py       # get_current_user + require_roles
      routers/auth.py       # login, register, forgot-password, reset-password
      routers/interviews.py # scheduling, join tokens, LiveKit token, signalling + transcript WS
      services/transcription.py  # ffmpeg -> AssemblyAI
      services/livekit_tokens.py # mint LiveKit join tokens
      services/interview_scheduling_queue.py # emails the manager when score > 65
      models/password_reset.py   # reset-token table
  frontend/
    nginx.conf.template     # /api proxy + WebSocket upgrade + CSP + no-cache
    index.html              # home/auth/workspace (+ Forgot password link)
    interview.html          # the consent LOBBY
    meeting-room.html       # the video room (icon controls)
    forgot-password.html / reset-password.html
    js/api.js               # same-origin /api + auth-expiry redirect
    js/interview-lobby.js / js/meeting-room.js
    js/livekit-client.umd.min.js  # bundled LiveKit SDK
```